

Radix-4 Booth, 2 PPs per slice

an 8-bit signed B split into two nibbles, recoded across the nibbles

Step 1 — split B into two nibbles

H is the value of the high nibble read as signed, L the value of the low nibble read as unsigned, b_3 the top bit of L :

$$B = 16H + L \quad H \in [-8, 7] \quad L \in [0, 15]$$

Today: H is signed and takes 2 Booth digits, L is unsigned and takes 3.

Step 2 — read the low nibble as signed

Booth reads every nibble as signed, so the low nibble comes out with its top bit counting -8 instead of $+8$, that is $16b_3$ too small:

$$L' = L - 16b_3$$

Two digits, no third one. On its own L' is wrong by $-16b_3$.

Step 3 — give the missing $16b_3$ to the high nibble

Booth's first window of the high nibble has a "bit below". Feed it b_3 instead of 0: that adds $+1$ to the high nibble, i.e. $+16$ to B , exactly when $b_3 = 1$.

$$H' = H + b_3 \quad 16H' + L' = 16H + 16b_3 + L - 16b_3 = B$$

H' can be 8, one more than a nibble holds, but two Booth digits reach 10. Four digits for B , two per nibble; the bit below is 0 for the lowest nibble or a whole 4-bit operand, and the top bit of the nibble below otherwise.

Step 4 — the dot product

The low nibbles go to one DP8, the high nibbles to another; the tree adds the high DP8 shifted by 16:

$$\sum_i A_i B_i = 16 \underbrace{\sum_i A_i H'_i}_{\text{high DP8}} + \underbrace{\sum_i A_i L'_i}_{\text{low DP8}}$$

Each DP8 alone is off by the same term $E = \sum_i b_{3,i} A_i$, with opposite signs and weights:

$$\text{low DP8} = \sum_i A_i L_i - 16E \quad \text{high DP8} = \sum_i A_i H_i + E$$

The $\times 16$ of the tree turns $+E$ into $+16E$ and cancels the $-16E$.

Step 5 — adding several DP8s first

Adding all low DP8s together, all high DP8s together, and then the two sums:

$$\left[\text{low}_1 + \text{low}_2 \right] + 16 \left[\text{high}_1 + \text{high}_2 \right] = \sum_i A_i (L'_i + 16 H'_i) = \sum_i A_i B_i$$

The error is $-16 b_{3,i} A_i$ on the low side and $+16 b_{3,i} A_i$ on the high side for every element i , so it cancels whatever the grouping, as long as both nibbles of each element end up in the same sum with the high one weighted 16.